Prob. 7. (2&T. Ch. 8, 1.5)

Let $\langle \xi_1, \xi_2, \dots \xi_n, -1 \rangle$ be a characteristic for the wave eq-n. We would like to show that

(a) in $\mathbb{R}^{n+1}$ the planes,

(1) $$\xi_1 x_1 + \dots + \xi_n x_n - t = C$$

are characteristic surfaces with r.t. the wave eq-n:

(2) $$\frac{\partial^2 u}{\partial x_1^2} + \dots + \frac{\partial^2 u}{\partial x_n^2} - \frac{\partial^2 u}{\partial t^2} = 0$$

(b) Show that the plane wave sol-ns

(3) $$u(x_1, x_2, \dots x_n, t) = F(\xi_1 x_1 + \dots + \xi_n x_n - t)$$

are constant on these planes.

Sol-n:

(a) [Preamble].

Consider the characteristic direction of the PDE in (2), using its defin-n:

(4) $$\xi_1^2 + \dots + \xi_n^2 - \xi_t^2 = 0,$$

if we let $\boxed{\xi_t^2 = 1}$, then

(5) $$\xi_1^2 + \xi_2^2 + \dots + \xi_n^2 - 1 = 0 \Rightarrow$$ Sol-n for this (5) is

some numbers $\{\xi_1, \xi_2 \dots \xi_n\}$. Then, the characteristic directions are

(6) $$\underline{M}_1 = \langle \xi_1, \xi_2 \dots \xi_n, \overset{\to \xi_t}{1} \rangle, \text{ or } \underline{M}_2 = \langle \xi_1, \xi_2 \dots \xi_n, -1 \}$$

Yet, the normal vector to the given plane in (1) is

(7) $$\underline{n} = \langle \xi_1, \xi_2, \dots \xi_n, -1 \rangle$$

which is parallel to both $\underline{M}_1$ & $\underline{M}_2$ (indeed $\underline{n} = \underline{M}_2$), hence the plane in (1) is the surface (charcter. surf.) where the information propogates. ▨

(b) By part (a), the plane in (1) is

(1) $$\xi_1 x_1 + \dots + \xi_n x_n - t = c,$$

so the wave sol-ns

$$u(x_1, x_2, \dots x_n, t) = F(\xi_1 x_1 + \xi_2 x_2 + \dots + \xi_n x_n - t) \overset{\text{by (1)}}{=} F(c) \Rightarrow$$

(8) $$u = F(c) = \text{const},$$

establishing the desired result. ∎

Prob. 8. (ZST, Ch. 8. 1.6)

Rewrite the spherical wave eq-n (why is this called the spherical wave eq-n?):

(1) $$\frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \cdot \frac{\partial u}{\partial r}\right] - \frac{\partial^2 u}{\partial t^2} = 0, \qquad (n=3)$$

as a one dim-l wave eq-n: $w_{rr} - w_{tt} = 0$ using the change of <u>dependent</u> variables $w = ru$.

Sol-n: [Preamble].

Consider the ↓ change of variables for dep-t var-le:

(2) $$w = r \cdot u \Rightarrow \boxed{u = \frac{w}{r}}, \quad \boxed{w = w(r,t)}. \text{ Then}$$

(3) $$\frac{\partial u}{\partial r} = \frac{\partial}{\partial r}\left[\frac{1}{r} \cdot w\right] = \frac{-1}{r^2} \cdot w + \frac{1}{r} \cdot w_r \quad \&$$

(4) $$\frac{\partial u}{\partial t} = \frac{\partial}{\partial t}\left[\frac{1}{r} \cdot w\right] = \frac{1}{r} w_t$$

Putting this into the LHS of (1) gives

$$\text{LHS} = \frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \cdot \left(-\frac{1}{r^2} w + \frac{1}{r} \cdot w_r\right)\right] - \frac{\partial}{\partial t}\left[\frac{1}{r} \cdot w_t\right] =$$